

Customer Insights Analytics Platform

Outcome: An end-to-end SaaS customer analytics stack — curated SQL/Python marts, a trained Gradient Boosting churn model (ROC-AUC 0.824), and two multi-page Power BI executive reports for retention/growth and support/SLA — with JSONPlaceholder engagement ingest and KPI HTTP sinks.

1. Why

SaaS operators need one customer view: who is growing, who is at risk of leaving, and how support and SLA performance affect the book. Without curated marts and scored churn risk, retention and support teams operate from disconnected extracts. This system closes that gap with a reproducible path from raw events to marts, ML scoring, and Power BI pages that rebuild from committed mart CSVs — no cloud credentials required for local validation.

2. What — Deliverables

- **Source extracts** — 5,000 customers; multi-year transactions (~81k rows) and support events (~8.5k) with SLA clocks and CSAT.
- **Curated marts** — customer_360, retention_monthly, revenue_cohorts, mrr_movement, segment_performance, product_revenue, support_sla, engagement_events.
- **Churn ML** — GradientBoostingClassifier with scaled numerics + one-hot categoricals; metrics, feature importance, holdout predictions, joblib artifact.
- **Power BI pack** — Retention & Growth (overview, cohorts, growth) and Support & SLA (overview, SLA, pending) with screenshots, semantic model, and DAX.
- **Warehouse SQL** — Snowflake-flavored *sql/marts/* matching local Parquet/CSV grains.
- **API source** — JSONPlaceholder users/posts/comments → engagement_events enrichment mart.
- **API sinks** — POST retention/support KPI snapshot to local FastAPI and JSONPlaceholder.
- **One-command rebuild** — *python run_pipeline.py* regenerates raw → marts → model → screenshots.

3. When — Delivery Plan

Phase	Focus	Typical duration
Design	Customer 360 grain, cohort/MRR definitions, churn label & features, Power BI page briefs	3–5 days
Build	Mart builders + SQL parity, churn training pipeline, Power BI model/DAX, API source/sinks	2–3 weeks
Validate	Metric reconciliation, holdout + CV review, report rebuild from mart CSVs, sink receipts	3–5 days
Handoff	EVALUATION.md, semantic model docs, runbook, acceptance review	2–3 days

Comparable scoped delivery: roughly **4–6 weeks**, with extra time if live warehouse credentials and Desktop report sign-off are in scope.

4. How — Architecture & Stack

SQL is the source of truth for warehouse grains; Python parity builders reproduce the same tables offline. The churn label correlates with tenure, usage, NPS, payment failures, and plan so the model is genuinely predictive. The stack runs fully offline for rebuilds; API stages are optional enrichment and delivery proof.

Concern	Choice
Marts	Python builders + Snowflake-flavored SQL parity
ML	scikit-learn GradientBoostingClassifier
Features	Tenure, usage, NPS, tickets, payment failures, health score, plan/segment/region/channel, revenue aggregates
BI	Power BI — 2 reports × 3 pages; semantic model + DAX
API source	JSONPlaceholder (users, posts, comments)
API sinks	Local FastAPI /ingest + JSONPlaceholder /posts
Orchestration	run_pipeline.py staged CLI

5. Architecture Diagram

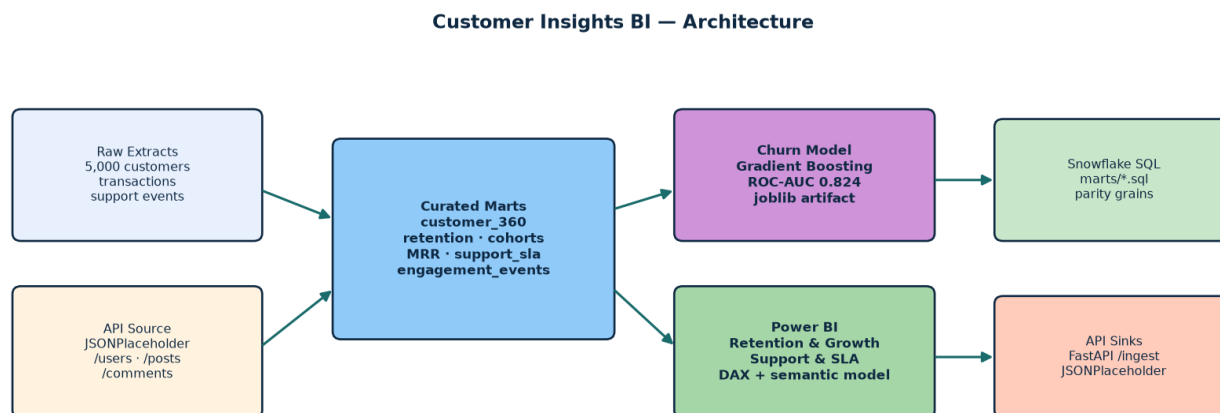


Figure 1. Raw + API → curated marts → churn model and Power BI; KPI sinks for downstream systems.

6. Technical Depth

Churn model evaluation (holdout + cross-validation)

Metric	Value
Model	GradientBoostingClassifier
Train / test split	3,750 / 1,250

Metric	Value
Test churn rate	12.2%
Accuracy	0.888
Precision / Recall / F1	0.597 / 0.243 / 0.346
ROC-AUC (holdout)	0.824
5-fold CV ROC-AUC	0.833 ± 0.018
Confusion matrix (test)	[[1073, 25], [115, 37]]

Top predictive features (importance)

Feature	Importance
tenure_days	0.252
txn_count	0.212
lifetime_revenue	0.151
customer_health_score	0.115
nps_score	0.057
feature_adoption_score	0.041
payment_failures_90d	0.040
monthly_active_days	0.036

Mart volumes (CSV grains)

Mart	Rows (approx.)
customer_360	5,000
engagement_events (API enrichment)	2,400
retention_monthly	899
revenue_cohorts	1,175
support_sla	8,504
mrr_movement / segment_performance / product_revenue	48 / 80 / 240

7. Outcomes — What the Client Gets

- A single customer 360 and retention/revenue/support mart suite ready for warehouse and BI.
- A trained, evaluated churn model with ROC-AUC ~0.82 and documented feature drivers.
- Two Power BI executive report packs with screenshots, DAX, and recreation briefs.
- Snowflake SQL scripts aligned to the same grains as local marts.
- Engagement API enrichment path and KPI HTTP sinks with landing receipts.
- Reproducible one-command pipeline for refresh and regression checks.

8. Repository

<https://github.com/user-JB007/customer-insights-bi>

License: MIT. No PII in committed extracts. See *artifacts/model/EVALUATION.md* and *docs/architecture.md* for full technical notes.